

Ray theory

From last lecture

Plane wave ansatz

$$u_i(x, t) = f(t - p_j x_j) a_i$$

Christoffel equation for plane waves:

$$\det[\Gamma(\mathbf{p}) - \mathbf{I}] = 0$$

$$\Gamma_{ik}(\mathbf{p}) = \frac{1}{\rho} A_{ijkl} p_j p_l$$

Definition of the slowness surface:

$$\det[\Gamma(\mathbf{p}) - \mathbf{I}] = 0$$



$$\begin{aligned}\nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} &= \mathbf{0}, & \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{H} + \frac{\partial \mathbf{D}}{\partial t} &= \mathbf{J}, & \nabla \cdot \mathbf{D} &= \rho,\end{aligned}$$

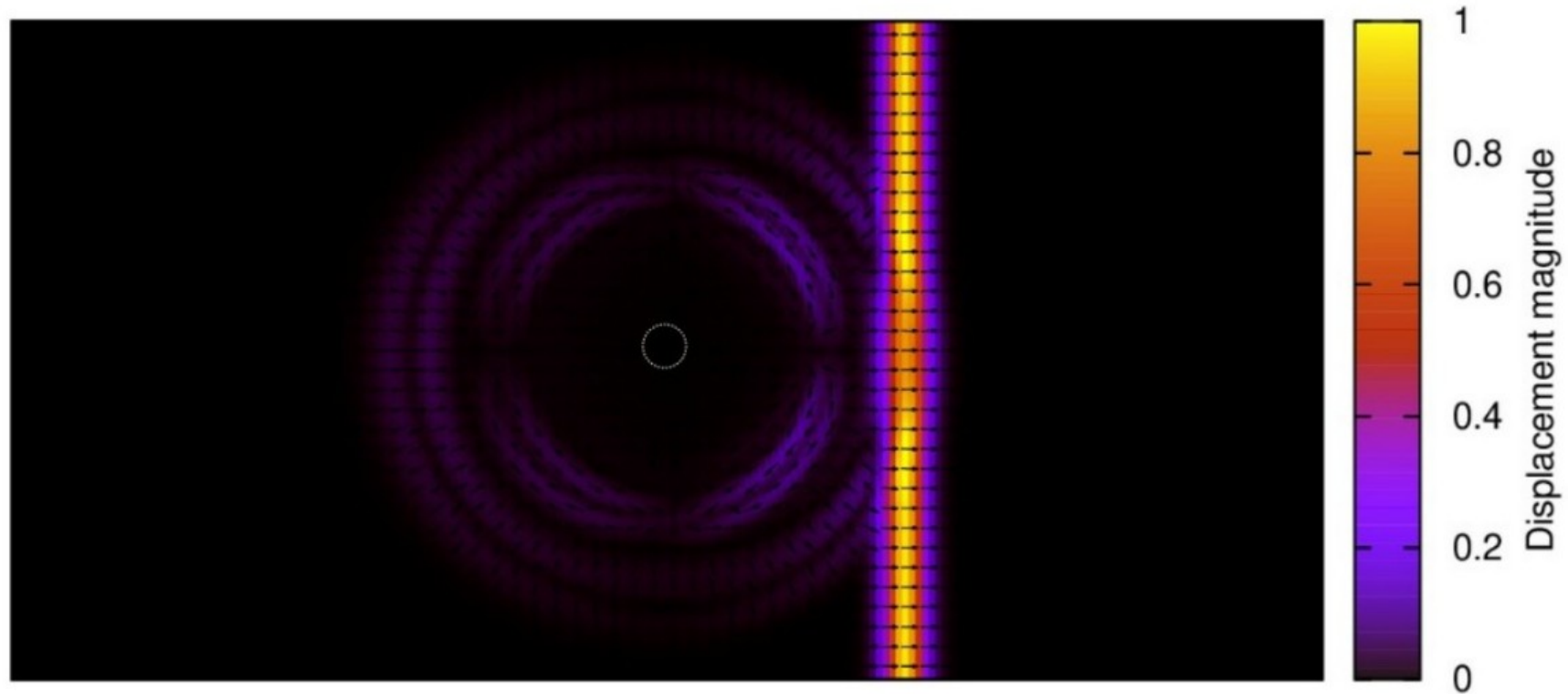
Quantum mechanics

$$i\hbar \frac{\partial \Psi}{\partial t} = H\Psi$$

Classical mechanics

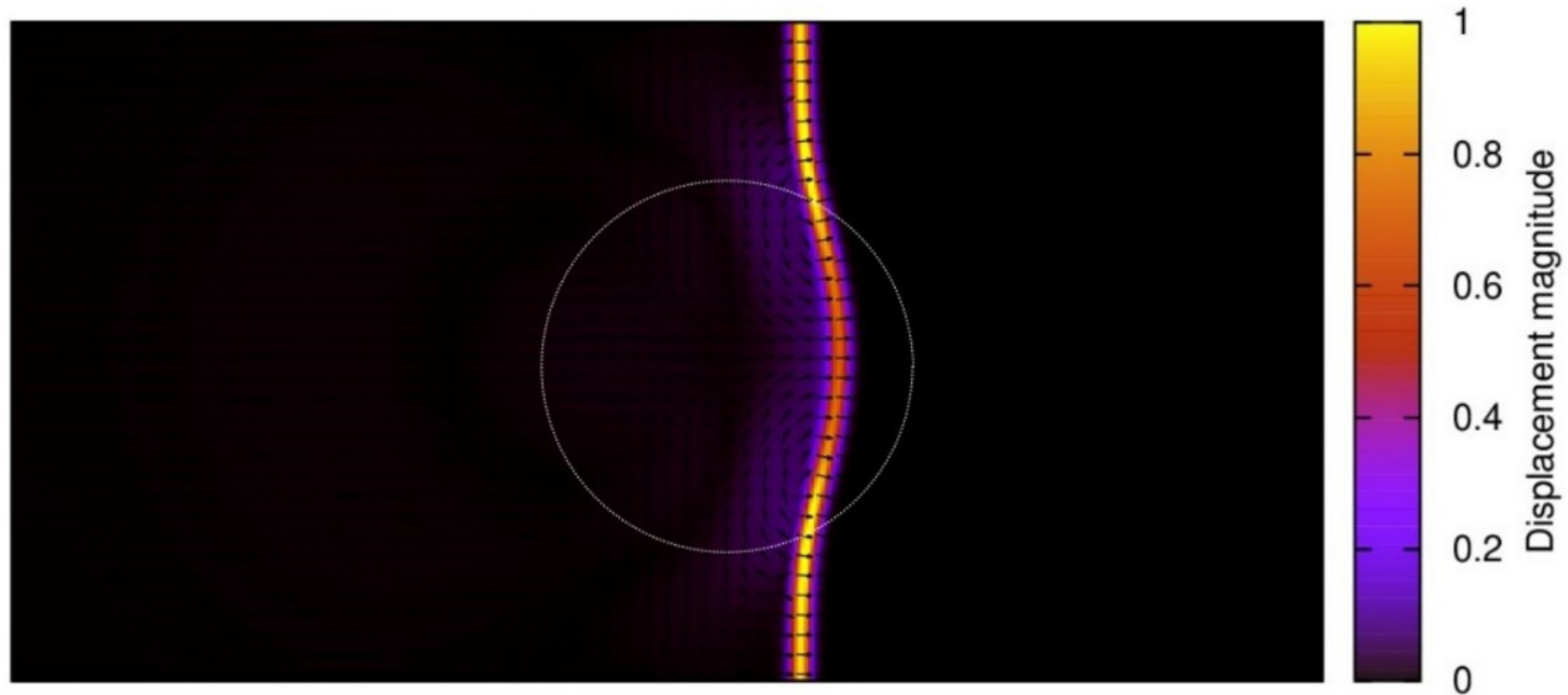
$$\begin{aligned}\frac{d\mathbf{q}}{dt} &= \frac{\partial H}{\partial \mathbf{p}} \\ \frac{d\mathbf{p}}{dt} &= -\frac{\partial H}{\partial \mathbf{q}}\end{aligned}$$

Length scales of heterogeneity



Scattering regime:

$$\frac{L}{\lambda} \approx 1$$

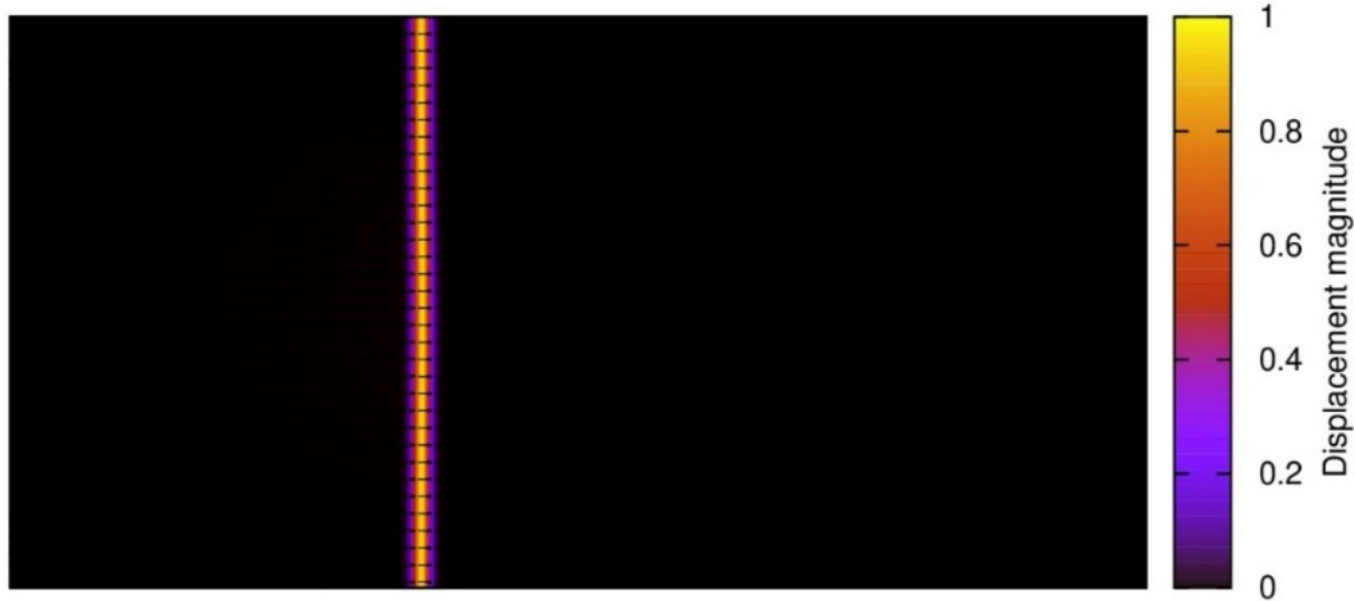


Geometric optics regime:

$$\frac{L}{\lambda} \gg 1$$

Ray series ansatz

Canonical problem

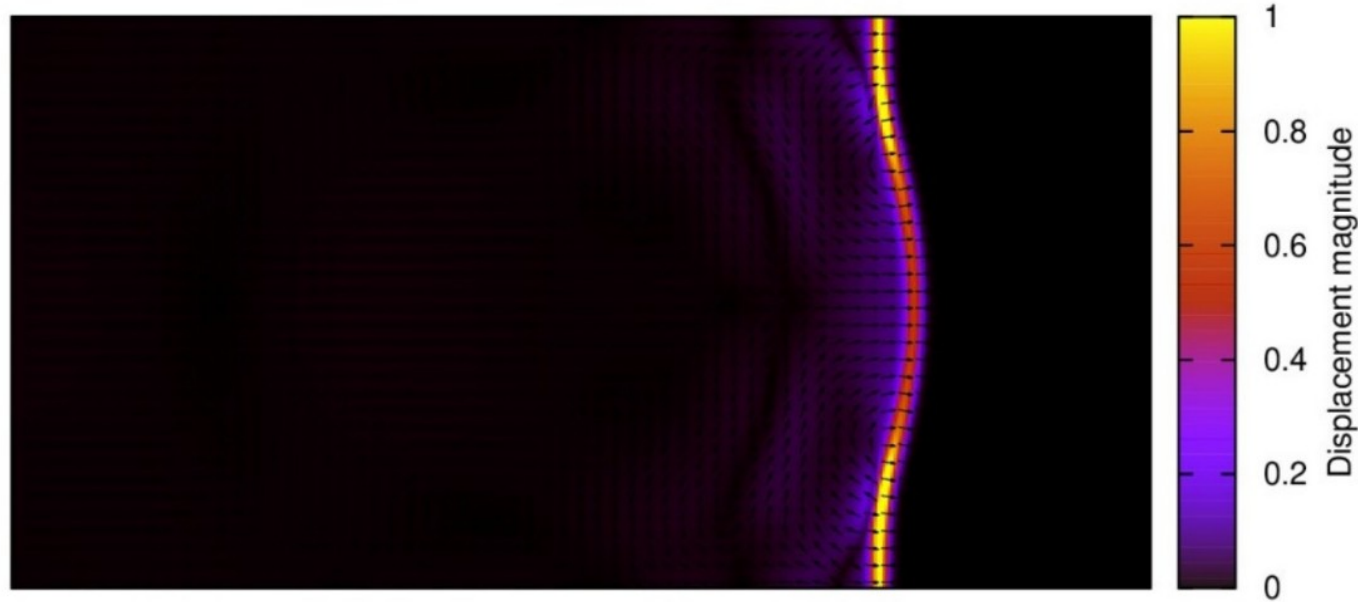


Start with a plane wave within the homogeneous left half plane:

$$u_i(x, t) = f(t - p_i^0 x_i) a_i^0$$

Can we extend this solution into the heterogeneous right half plane?

Initial ansatz



As an initial guess, we look for a solution of the form:

$$u_i(x, t) = f[t - T(x)]a_i(x)$$

Travel time and local slowness

Assumed solution takes the form:

$$u_i(x, t) = f[t - T(x)]a_i(x)$$

Introduction of the **travel time** function allows for curved wavefronts:

$$T(x) = \text{constant.}$$

The **local slowness vector** is orthogonal to the wavefronts:

$$p_i = \frac{\partial T}{\partial x_i}$$

Polarisation vector allowed to vary with position – accounts for **focusing effects** and for variations in anisotropic properties with position

The ray series

Harmonic decomposition solution:

$$u_i(x, t) = \frac{1}{2\pi} \int_{\mathbb{R}} \tilde{f}(\omega) e^{i\omega[t-T(x)]} a_i(x) \, d\omega,$$

This is a “summation” over singular solutions:

$$u_i(x, t) = \delta[t - T(x)] a_i(x)$$

Experience has shown that we should generalise this solution to become:

$$u_i(x, t) = \frac{1}{2\pi} \int_{\mathbb{R}} \sum_{n=0}^{\infty} \frac{a_i^n(x)}{(i\omega)^n} e^{i\omega[t-T(x)]} \, d\omega,$$

First term (n=0) in the **ray series** is the singular one, while successive ones become smoother and smoother

The eikonal equation

Frequency-domain equations

Linearised equations with no force term:

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = 0,$$

Transforming to the frequency-domain:

$$-\omega^2 \rho u_i - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = 0$$

Ray series in the frequency-domain:

$$u_i(x, \omega) = \sum_{n=0}^{\infty} \frac{a_i^n(x)}{(i\omega)^n} e^{-i\omega T(x)}.$$

We can now substitute in our ansatz and see what happens!

Lots of differentiation...

Example step from the calculation:

$$\frac{\partial u_k}{\partial x_l} = \sum_{n=0}^{\infty} \frac{1}{(i\omega)^n} \left(\frac{\partial a_k^n}{\partial x_l} - i\omega p_l a_k^n \right) e^{-i\omega T},$$

Factor of frequency is added each time we differentiate the phase term

Result of the calculation

Equations of motion imply:

$$\sum_{n=0}^{\infty} \frac{1}{(i\omega)^n} \left[\frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial a_k^n}{\partial x_l} \right) - i\omega \frac{\partial}{\partial x_j} (A_{ijkl} p_l a_k^n) \right. \\ \left. - i\omega A_{ijkl} p_j \frac{\partial a_k^n}{\partial x_l} + (i\omega)^2 A_{ijkl} p_j p_l a_k^n - (i\omega)^2 \rho a_i^n \right] e^{-i\omega T} = 0.$$

Common phase term cancels

These equations must hold for **all frequencies** – this implies that the **coefficients of each power of frequency must separately vanish**

Compare with power series solution of ODEs

Results of the calculation

From the highest order coefficient, we have:

$$[A_{ijkl}p_jp_l - \rho\delta_{ik}]a_k^0 = 0,$$

At the next order:

$$[A_{ijkl}p_jp_l - \rho\delta_{ik}]a_k^1 = \frac{\partial}{\partial x_j}(A_{ijkl}p_la_k^0) + A_{ijkl}p_j\frac{\partial a_k^0}{\partial x_l},$$

For the general term with $n > 1$:

$$[A_{ijkl}p_jp_l - \rho\delta_{ik}]a_k^n = \frac{\partial}{\partial x_j}(A_{ijkl}p_la_k^{n-1}) + A_{ijkl}p_j\frac{\partial a_k^{n-1}}{\partial x_l} - \frac{\partial}{\partial x_j}(A_{ijkl}\frac{\partial a_k^{n-2}}{\partial x_l})$$

First equation determines **p**, while the later ones successively determine the polarisation vectors at different orders

Where have we made an approximation?

A short cut

In this course we will consider only the equation derived from the **highest-order** term in the expansion

If that is all that we required, we can systematically drop all but the highest order terms within the previous derivation:

$$\frac{\partial u_k}{\partial x_l} = \sum_{n=0}^{\infty} \frac{1}{(i\omega)^n} \left(\frac{\partial a_k^n}{\partial x_l} - i\omega p_l a_k^n \right) e^{-i\omega T} \approx -i\omega p_l a_k^0 e^{-i\omega T}$$

This saves a lot of time!

You can practice these steps in Example sheet 3

Highest order condition

From the highest order coefficient, we have:

$$[A_{ijkl}p_jp_l - \rho\delta_{ik}]a_k^0 = 0,$$

This motivates the definition of a local Christoffel matrix:

$$\Gamma_{ik}(\mathbf{x}, \mathbf{p}) = \frac{1}{\rho(\mathbf{x})} A_{ijkl}(\mathbf{x}) p_j p_l.$$

And so write the above condition as:

$$\{\Gamma[\mathbf{x}, \mathbf{p}(\mathbf{x})] - \mathbf{I}\} \mathbf{a}^0(\mathbf{x}) = \mathbf{0},$$

Locally, things look identical to plane waves!

The eikonal equation

The equation

$$\{\Gamma[\mathbf{x}, \mathbf{p}(\mathbf{x})] - \mathbf{I}\} \mathbf{a}^0(\mathbf{x}) = \mathbf{0},$$

has non-trivial solutions for the zeroth-order polarisation vector if and only if:

$$\det\{\Gamma[\mathbf{x}, \mathbf{p}(\mathbf{x})] - \mathbf{I}\} = 0,$$

This is a non-linear first order PDE for the travel time called the **eikonal equation**

If the local slowness surfaces are everywhere non-degenerate:

$$||\mathbf{p}(\mathbf{x})||^2 c_k[\mathbf{x}, \hat{\mathbf{p}}(\mathbf{x})]^2 - 1 = 0$$

The method of characteristic

Ray Hamiltonian

Recall the eikonal equation is:

$$||\mathbf{p}(\mathbf{x})||^2 c_k[\mathbf{x}, \hat{\mathbf{p}}(\mathbf{x})]^2 - 1 = 0$$

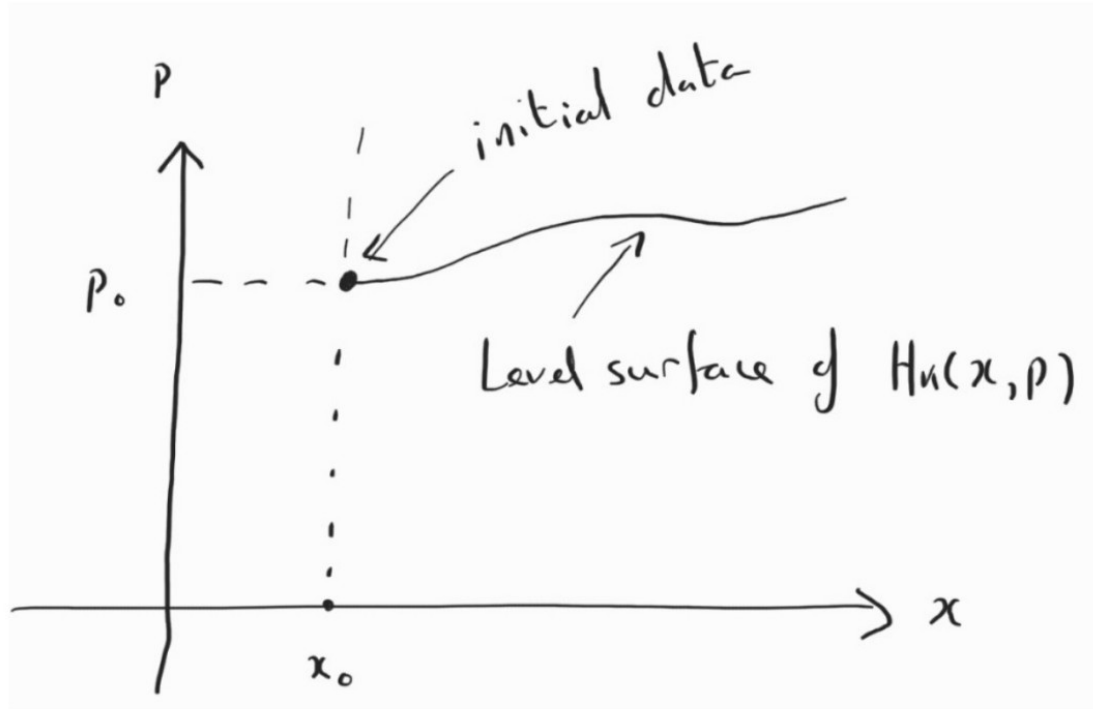
If we defined the **ray Hamiltonian** for the k th sheet of the slowness surface:

$$H_k(\mathbf{x}, \mathbf{p}) = \frac{1}{2} ||\mathbf{p}(\mathbf{x})||^2 c_k[\mathbf{x}, \hat{\mathbf{p}}(\mathbf{x})]^2,$$

We can write the eikonal equation in the form:

$$H_k[\mathbf{x}, \mathbf{p}(\mathbf{x})] = \frac{1}{2}, \quad \mathbf{p}(\mathbf{x}) = (\nabla T)(\mathbf{x}).$$

Geometric interpretation



Solution of eikonal equation is a 3D level surface of the Hamiltonian in a 6D phase space

Initial data specifies a 2D level surface in the phase space

The problem is how to extend this 2D surface into the appropriate 3D level surface

This is where Hamilton's equations enter into the method

Hamilton's equations

Initial data extended through integration of Hamilton's equations:

$$\frac{dx_i}{d\sigma} = \frac{\partial H_k}{\partial p_i}, \quad \frac{dp_i}{d\sigma} = -\frac{\partial H_k}{\partial x_i}$$

$$x_i(0) = x_i^0, \quad p_i(0) = p_i^0.$$

This works because the Hamiltonian is constant along the integral curves:

$$\frac{d}{d\sigma} H[\mathbf{x}(\sigma), \mathbf{p}(\sigma)] = \frac{\partial H}{\partial x_i} \frac{dx_i}{d\sigma} + \frac{\partial H}{\partial p_i} \frac{dp_i}{d\sigma} = -\frac{dp_i}{d\sigma} \frac{dx_i}{d\sigma} + \frac{dx_i}{d\sigma} \frac{dp_i}{d\sigma} = 0,$$

There are a few subtler points to the method that are being ignored

Recovering the travel time

Rays are the projection of integral curves into the **position-domain**

Rate of change of the travel time along a ray:

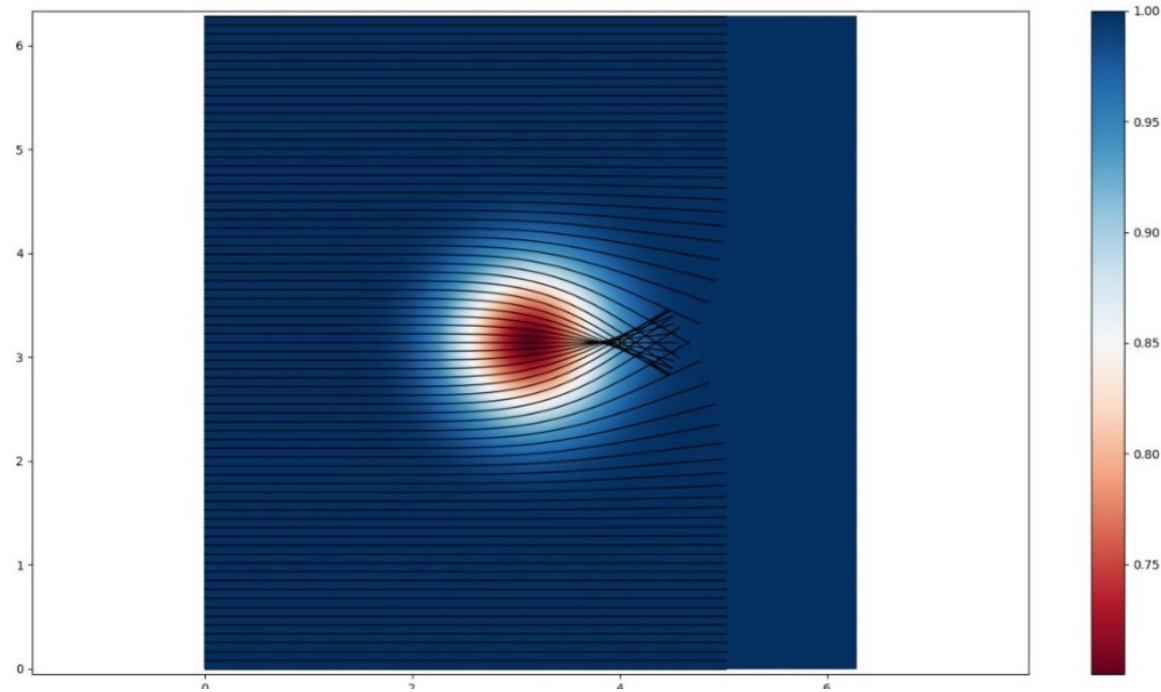
$$\frac{d}{d\sigma} T[\mathbf{x}(\sigma)] = \frac{\partial T}{\partial x_i} \frac{dx_i}{d\sigma} = p_i \frac{\partial H}{\partial p_i}$$

Right hand side simplified using Euler's homogeneous function theorem;

$$\frac{d}{d\sigma} T[\mathbf{x}(\sigma)] = 1.$$

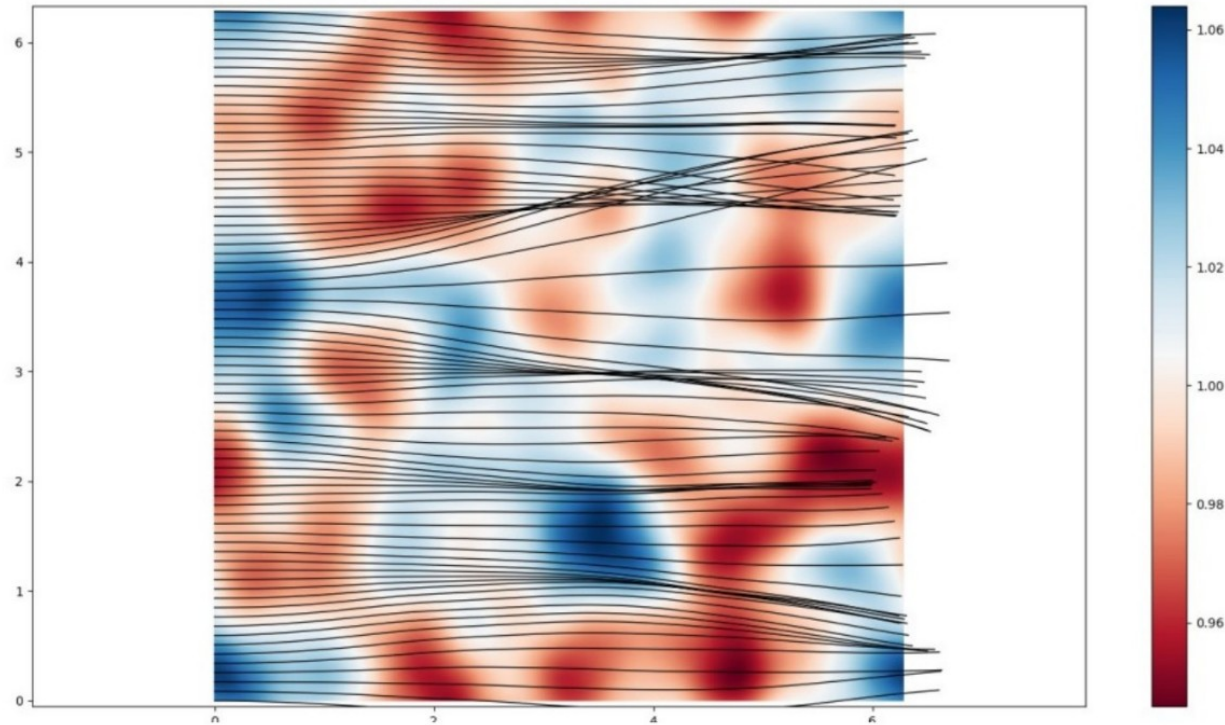
The generating parameter along the ray is equal to the travel time

Caustics



If rays for two different integral curves touch, then the travel time need not be well-defined. These points are known as **caustics**. The **ray theoretic amplitude is infinite** at such points, while the true amplitude is large but finite

Caustics



Within heterogeneous materials, caustics occur frequently – c.f. Penrose Hawking singularity theorems in general relativity